Systems and Devices 3 (SYS3)

Lecture Review – Week 3

Dr Pengcheng Liu

Week 3 Module
Overview and Updates

Module Overview and Updates

- **Week 3**:

Lecture: this week, our discussion was focused on some principles of the RISC Pipeline. Also, we had our first Tutorial Lecture on Performance.

Lab: In the lab, we started exploring the idea of structural hazards in a CPU pipeline, and possible solutions. You started with some basic ideas, then develop an algorithm to solve structural hazards in the pipeline. In Week 4 an addittional help sheet will be released to assist those who have not been able to devise a suitable idea to implement. Solutions to algorithm problems will be released at a later date.

Exercises/Quizzes: Quiz 1 was release on 20/02/2026. This is a formative quiz based on week 1, week 2 and week 3 lectures, responses will be automarked. You can try as many times as you like till you get a perfect score. Quiz 1 will be discussed in the Lecture Review session on 12/03/2026 (Week 5 Thursday).

Module Overview and Updates

- **Next Week (Week 4)**:

Lecture: Next week, our discussion will be focused on Pipeline Hazards. Lecture 6 and Lecture 7 will be discussing on three major types of pipeline hazards: structural hazard, data hazard and control hazard. Lectures will release tomorrow. The questions for Tutorial Lecture TL2 will also be released. You may want to attempt to solve them before watching the Tutorial Lecture TL2 video.

Lab: In Week 4, you should be continuing work from Week 3, to complete the development of an algorithm to resolve structural hazards, and test it. This week a help sheet is released to allow students to follow a suggested approach for those who have not yet worked out an idea for a solution.

Exercises/Quizzes: Solutions for Exercise 1 will be release on 06/03/2026. You are expected to complete Exercise 1 by this date. Exercise 1 will be discussed in the Lecture Review session on Thu 12th March (S2 Week 5).

Week 3

Guest Lecture

Speaker: Dr Wei Wu

Title: RISC-V: A Brief History and Roadmap to 2036

Location: PZA/103 Lecture Theatre, Campus East - Piazza Building

On Zoom https://york-ac-uk.zoom.us/j/91407678535?pwd=qapZLiRTCcI7DlNIHTPOQNk62UypyN.1&jst=2

Time: Thursday, 26 February·9:00 – 11:00am

Abstract: As a young and open ISA standard, the RISC-V has the potential to capture a quarter of the global processor market share within its 20-year lifespan. Some bold predictions even suggest that RISC-V will become the most dominative ISA in the worldwide IT industry by 2036, serving as a cornerstone of the global open-source ecosystem due to its open and free nature.

In this lecture, the speaker will begin with the historical context of RISC-V's emergence, explaining the historical opportunities and inevitability of its creation as an open and free ISA. Furthermore, the speaker will showcase the remarkable achievements RISC-V has made in the software and open-source chip ecosystems, particularly the policy support from EU, China, Brazil, and India.

Week 3

Guest Lecture

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Title: RISC-V: A Brief History and Roadmap to 2036

Biography: Mr. Wei Wu is the founder and director of Project Jiachen, whose mission is to make RISC-V the most dominant instruction set architecture for all IT scenarios before 2036, achieving commercial prosperity. Currently, more than 50 companies, research institutions, and developers worldwide have joined the project, covering almost all major RISC-V IP companies. To achieve this vision, he founded PLCT Lab in 2019 and serves as its project director. Improving the RISC-V software ecosystem has been set as one of PLCT lab's most important strategic objectives. His team maintains a series of key open-source infrastructures, including OpenJDK RISC-V backend, V8 RISC-V backend, Spidermonkey RISC-V JIT backend, LuaJIT RISC-V downstream implementation, as well as GNU Toolchain and Clang/LLVM RISC-V backend. The TARSIER project, which he launched at the end of 2021, has taken a significant step forward in RISC-V desktop computing and Linux distribution support. In October 2025, he joined KUBUDS, a company he co-founded in 2023, to help RISC-V companies achieve greater success in the commercial industrial software ecosystem.

Wei has served as a director of the LLVM Foundation Board of Directors (BoD) since 2022. He is very passionate about educating compiler techniques and RISC-V to engineers and amateurs. He often gives talks and tutorials among amateurs and college students in local meetups. He also collaborates with college teachers, providing RISC-V software lectures and experiment courses.

Thanks

See You Next Week !